

Portfolio Construction

conflicting objectives $\begin{cases}$ security + stability of costs (matching)
high long term invest. returns (+ costs) $\end{cases}$

STEP 1 Establish appropriate asset mix for the fund
→ **strategic benchmark**

→ **Strategic (policy) risk**

STEP 2 tactical implementation of strategy by selecting one or more managers and decision on appropriate level of risk these managers should take relative to the benchmark

→ **Active (manager) implementation risk**

Strategic Risk - risk of poor performance of strategic benchmark relative to value of liabilities

Active Risk - risk of active manager underperforming the strategic benchmark

Structural Risk - mismatch between aggregate of portfolio benchmarks + total fund benchmark

Total Risk = Strategic Risk + Active Risk + Structural Risk

CHAPTER 22: PORTFOLIO MANAGEMENT

Multifactor - model

Attempt to model historic returns based on systematic risk factors

$$R_i = \alpha_i + b_{i1}I_1 + \dots + b_{iL}I_L + \epsilon_i$$

- R_i = return
- α_i, ϵ_i = random constants
- b_{ik} = sensitivity of factor i to factor k
- $I_k = \Delta$ in set of L factors explain variation in R_i around α_i

Active management	Passive management
① Estimate $E(\text{return})$ on a share given a set of risk factors + estimated factor returns - $E(\text{value})$ for each factor - substitute into model to predict $E(\text{return})$ on share ② Estimate return based on relationship $\begin{cases}$ returns sources of systematic risk$\end{cases}$ R SSR	Identify + control Exposure of particular portfolio to: - different risk factors - Δ in risk profile of portfolio to better match exposure at L Stratified sampling → shares Synthetic fund approach → cash + derivatives

↳ actively calculate Δ ID + control exposure

Technical Analysis

③ short-term price movements

Estimate future price + yields based on past prices, yields + trading volume;

→ use to predict $\begin{cases}$ trading ranges
sell signals
buy signals $\end{cases}$

Methods

1. Chartism

- trends + patterns
- links repeatability of patterns to investor psychology
- predict future movements

MTL support resistance } buy + sell signals < round number patterns of past movements

2. Mechanical trading rules

- removes subjectivity
- signals given on jet price movements
- too small → profits lost (trading expenses = Φ)
- too high → substantial price movement before trade

3. Relative Strength analysis

- performance vs market/sector
- mean reversion
- momentum

Advantages DETS	Disadvantages LAD
- easy to collect data - quick + easy to carry out - helpful → decisions on timing of investments - short term trading profits	- may distract investors from MB considerations ie. long-term value - could make hefty loss - may encourage more active trading = + expenses

Risk Budgeting

Establish how much investment risk should be taken + where is most efficient to take it in order to maximise returns

STEP 1: Decide how to allocate max permitted overall risk to

- total fund active risk
- strategic risk

STEP 2: Allocate total fund active risk budget across component portfolios

Summary of process:

- DEFINE** feasible set
 - assets to include
 - estimate volatilities + covariances
- CHOOSE** initial asset allocation using risk-return optimisation process + VaR to determine total risk tolerance
 - M-L modelling → determine matched portfolio
 - VaR calc → determine total risk budget
 - allocate risk budget → strategic + total active risk
 - model investment risks + returns → investment strategies
 - present results to clients → choose strategies
 - allocate active risk between different active managers
 - choose investment managers + allocate their benchmarks
- MONITOR** risk exposure + change in volatilities + correlations
- REBALANCE** portfolio in response to A in short-term conditional volatility + correlations of assets
 - keep overall portfolio risk at same tolerable level

 target risk level

- cost of investments
- diversification → reduces risk

Stress Test

- tailored to reveal weakness
- 1. risk exposure
- 2. sensitivity

VaR → **NOT** account for simultaneous + in return < volatility correlations observed during extreme market events

identify + investigate risks using stress testing

Portfolio

radically change underlying assumptions + characteristics to gain insight to sensitivity to predefined risk factors

Two types

- identify weak areas + investigate effects of localised stress situation by looking at effect of combo of S + corr
- gauge impact of major market turmoil affecting all model parameters while ensuring consistency between corr while 'stressed'

Measuring Risk

Tracking Error	error in following benchmark - used by positive managers who seek to follow index
Forward-looking tracking error	~ std dev (portfolio returns) that may be experienced in the future if current structure remains unchanged - quantitative modelling techniques - assumptions < volatility correlations
Backwards-looking tracking error	Annualised std dev (portfolio returns - benchmark returns) - observed relative performance
Active Money	deviation from benchmark portfolio for specified position in isolation $AM=0 \rightarrow$ fund run in perfect passive index-line fashion
Information Ratio	$IR = \frac{\text{mean (relative return)}}{\text{stddev (relative return)}}$ → relative return → historical tracking error how efficiently additional risk is converted into return as part of risk budgeting process
Downside Risk	std dev → looks at uncertainty investors dislike low returns ∴ identify worst period of underperformance in past - frequency - downside semi-stddev
Further risk adjusted measures	helps to decide if active risk = value
Active + Strategic Risk	Tracking Error $\approx 0.1 - 0.1$ Strategic risk → difficulty in L not < well defined objectively determined → market-related approach → take on more strategic risk
Value at Risk (VaR)	generalises likelihood of underperforming by providing a statistical measure of downside risk Asses: → LTC - potential loss - over given future time period - with given degree of confidence Assumptions: return $\sim N(\mu, \sigma)$ • not appropriate if < fat-tailed skewed

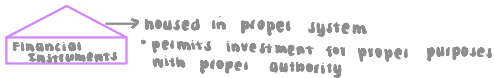
who exhibits non-normal distributions?

- portfolios with < credit risk systematic bias derivatives
 usefulness of VaR < Gumbel, Frechet, Weibull Monte Carlo simulations

Custodian

holds investment securely on behalf of investor

Principle function:



Services: **SCITCET**

- offer custody of documents
- income collection
 - collect coupons + dividends
 - chase missing payments
- tax recovery
 - withholding tax deducted but reclaimed
- cash management
 - manage to settle deals
 - account = sufficient funds to settle outstanding purchases
- securities settlement
 - main function
 - admin of actual exchange
- foreign exchange trading
 - settle deals
- stock lending
 - shares not traded → loaned
 - other company = short position

Global custodian → provide ^{1.} international services + responsibility for custody of ^{2.} assets held overseas

- may appoint sub-custodian
- superior local knowledge
 - comply with local regulations